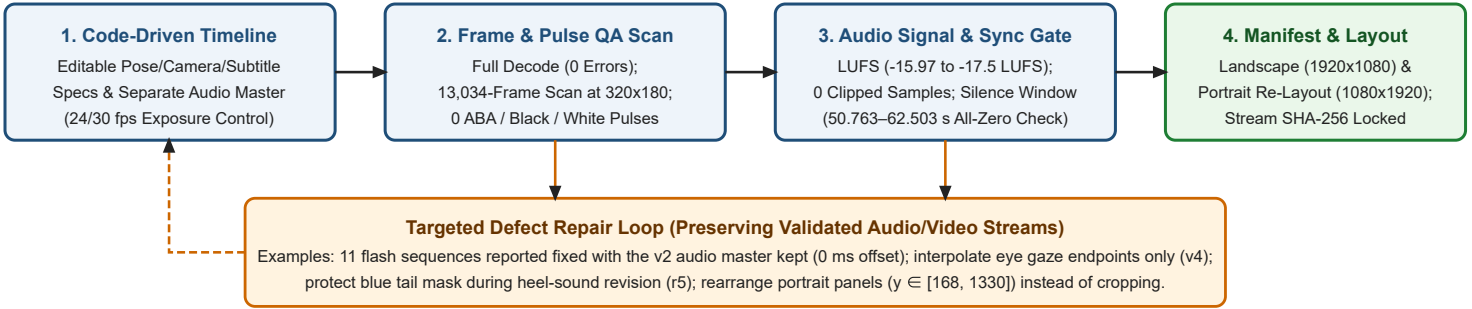

1. Code-Driven Timeline
Editable Pose/Camera/Subtitle Specs & Separate Audio Master (24/30 fps Exposure Control)
2. Frame & Pulse QA Scan
Full Decode (0 Errors); 13,034-Frame Scan at 320x180; 0 ABA / Black / White Pulses
3. Audio Signal & Sync Gate
LUFS (-15.97 to -17.5 LUFS); 0 Clipped Samples; Silence Window (50.763–62.503 s All-Zero Check)
4. Manifest & Layout
Landscape (1920x1080) & Portrait Re-Layout (1080x1920); Stream SHA-256 Locked
Targeted Defect Repair Loop (Preserving Validated Audio/Video Streams)
Examples: 11 flash sequences reported fixed with the v2 audio master kept (0 ms offset); interpolate eye gaze endpoints only (v4); protect blue tail mask during heel-sound revision (r5); rearrange portrait panels (y ∈ [168, 1330]) instead of cropping.